

Mapping and detecting ridges on metal coated products

Jan. 2025 – Sept. 2025 | BLUESCOPE STEEL

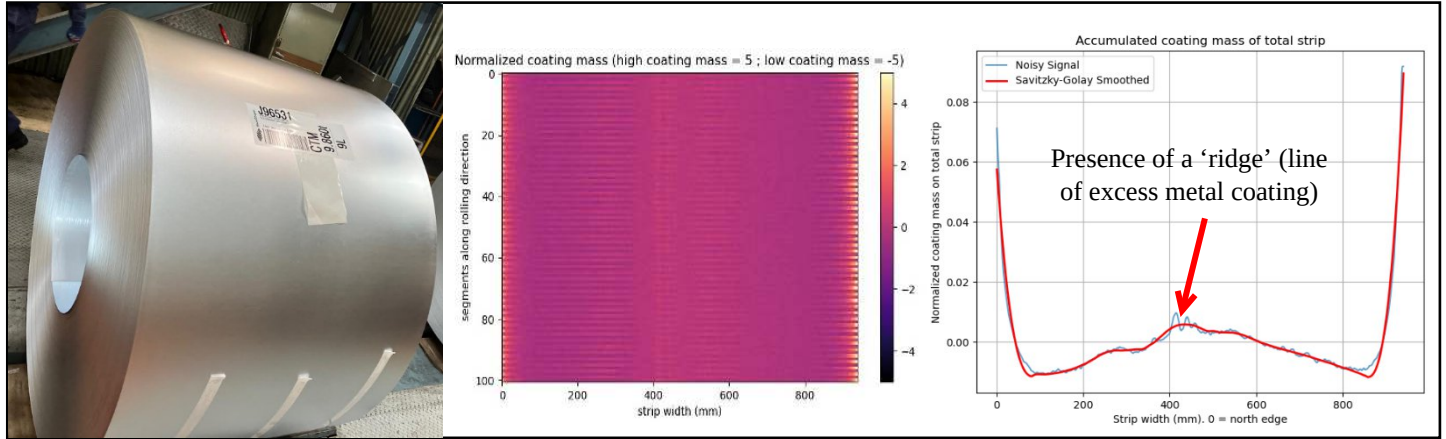


Figure 1: during hot dip galvanizing, the airflow used to control the amount of coating on the strip can be disturbed, and result in a line of excess coating in the rolling direction. Over multiple wraps, the regions with excess coating (a 'ridge') can deform the outer wraps of the coil. Using signal processing techniques and linear interpolation, data from the x-ray fluorescence coating weight gauge can be used to detect the presence of a ridge

Objectives and Motivation

Aims

- Main objective was to detect the formation of a ridge within a metal coated coil during line-run
 - *Defining a ridge → line of excess metal coating in the rolling direction of the strip, causing plastic deformation on the outer wraps of a coil. A ridge can also compromise the paint's adhesion to the metal surface. Excess coating can be caused by debris hindering the airflow from jets, which control the amount of metal coating stripped from the surface during hot dip galvanizing.*
- Data used to detect ridge formation would be taken from an XRF coating weight gauge, that measures the amount of metal coating on continuously moving strip

Motivation

- Ridges are subtle defects that are difficult to capture on a production line, and cause issues for downstream customers during painting and roll-forming of the product.
- Detecting the defect during line-run can avoid unknowingly generating defective product and thus consequently allow for immediate rectification by operations.

Results

Summary of algorithm used to filter out ridges

- Due to the incoming XRF data being filtered, the discrete inverse Fourier Transform is used to acquire the raw dataset
- The XRF data used to measure the top and bottom surfaces were individually added together using linear interpolation, where the net result would ideally show the formation of a ridge (area where there is excess metal coating). The colourmap illustrates this concept in Fig. 1.
- To automatically identify the peak where a ridge forms (plot in Fig. 1), the raw/noisy signal is subtracted from the smoothed data (obtained via a Savitsky Golay filter), where the remainder would show the amplified peak. This is a way for operators to be automatically alerted when a ridge is being formed.

Validation and implementation of algorithm onto production line

- Multiple ridge samples were taken from defective coils, and were analysed to verify the location of a ridge. This assisted in validating the position of the ridge predicted by the algorithm.
- Currently the algorithm is in the process of being implemented onto the production line by the automation engineering teams.

Defect detection in extruded polyethylene film for QA testing

Aug. 2023 – Nov. 2023 | QENOS

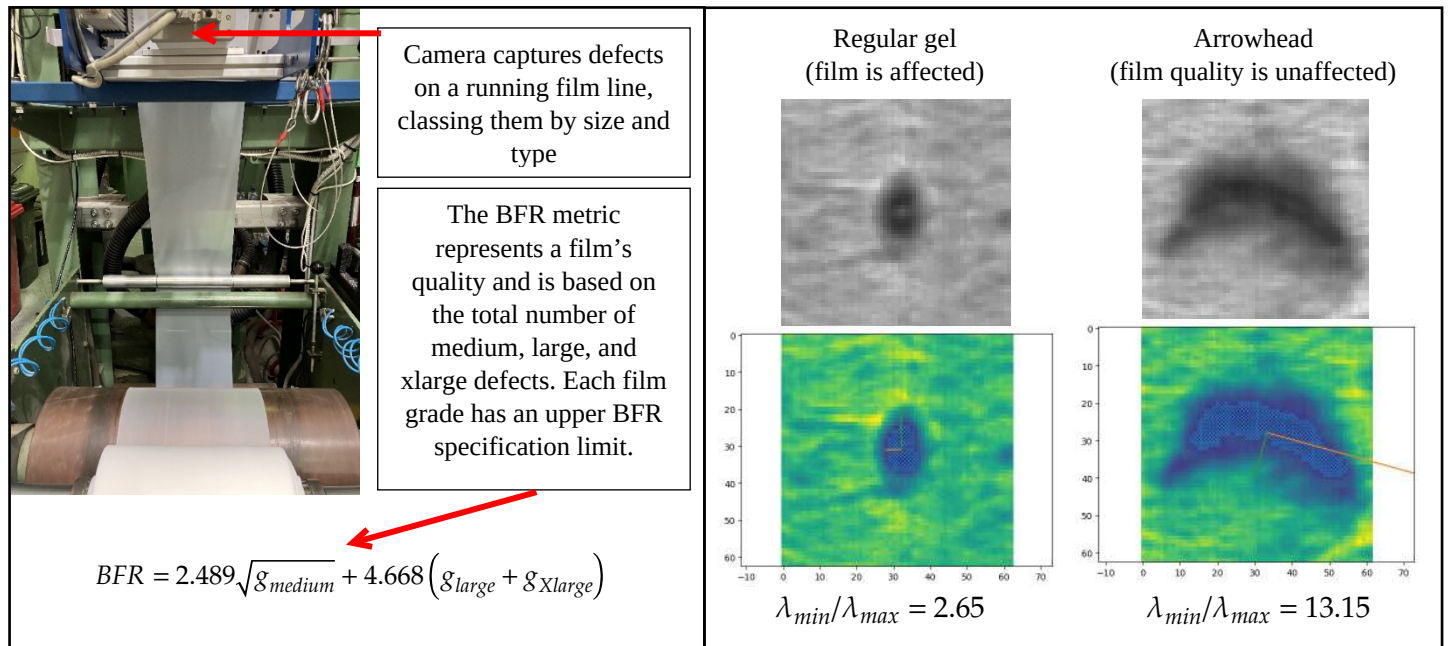


Figure 2: photo of a running film line extruding polyethylene. QA testers use the BFR value to quantify a film's aesthetic quality.

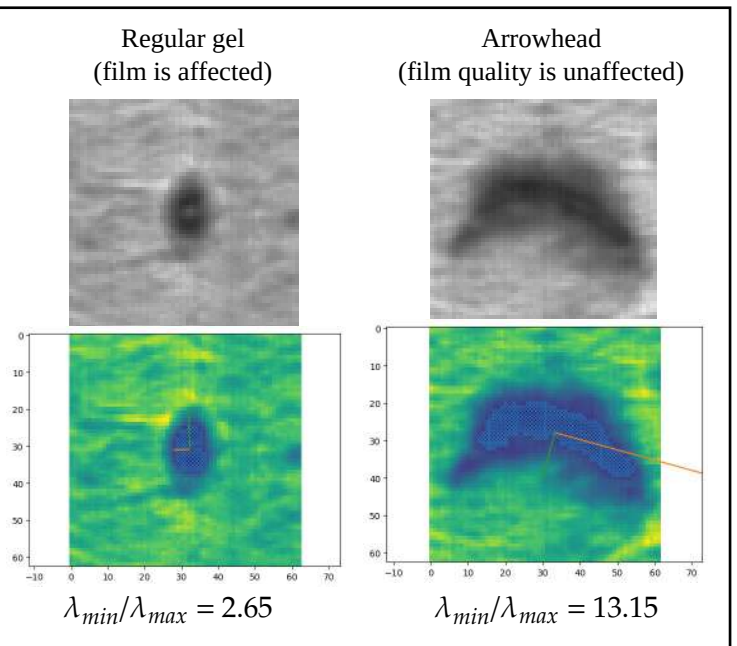


Figure 3: example of how principal component analysis was used to distinguish between gels (affects film quality) and arrowheads (does not affect film quality). Due to arrowheads having a higher aspect ratio than gels, the ratio of their min. and max. eigenvalues is greater.

Objectives and Motivation

Aims

- Main objective was to improve the reproducibility of current QA procedures for testing extruded polyethylene films.
- Localized defects called 'gels' form during film extrusion, which negatively impact both mechanical and visual performance.
- A camera photographs these defects on a running film line and categorises each defect by size (small, medium, large, xlarge) and type. Both categorisations result in the output of a single numeric (known as BFR) which summarises the film's quality (Fig. 2).
 - Large defects significantly increase the BFR, possibly giving off-specification results.

Motivation

- For a given batch the BFR fluctuated with time, giving variable results that are not representative of the film's quality.
- Thus, the gel data was dependent on time, which was not adequate for our QA gel testing.

Results

Reconfiguring camera settings

- Crescent-shaped defects called 'arrowheads' were falsely classified as gels. Previous studies showed arrowheads did not impact the visual and mechanical properties of the film.
- Due to their erratic presence during extrusion, this would contribute to fluctuations in the BFR.
- The camera was configured to exclude arrowheads from BFR calculations, based on rules related to its unique geometry.
- An algorithm (based on principal component analysis) was used to verify the accuracy of the camera's arrowhead detection (Fig. 3). Hence, the 'true/correct' classifications of arrowheads were done by the PCA algorithm, and so the camera's probability for successfully detecting an arrowhead was 86%.

Impact of temperature on polymer processing

- Regular gel counts were found to increase with time, in addition to arrowhead contamination.
- From rheological testing, it was found high-flow polymers would accumulate within the walls of the extruder and eventually flush out as gels. This in turn would contaminate the film, giving a high BFR.
- Hence, reducing the processing temperature lowered the polymer's MI, which gave more reproducible results as the gel counts were consistent with time. This also reduced the presence of arrowheads.

Determination of strain hardening modulus in relation to slow crack growth for polyethylene (PE) materials in piping systems (ISO:18488)

Aug. 2023 – Jan. 2024 | QENOS

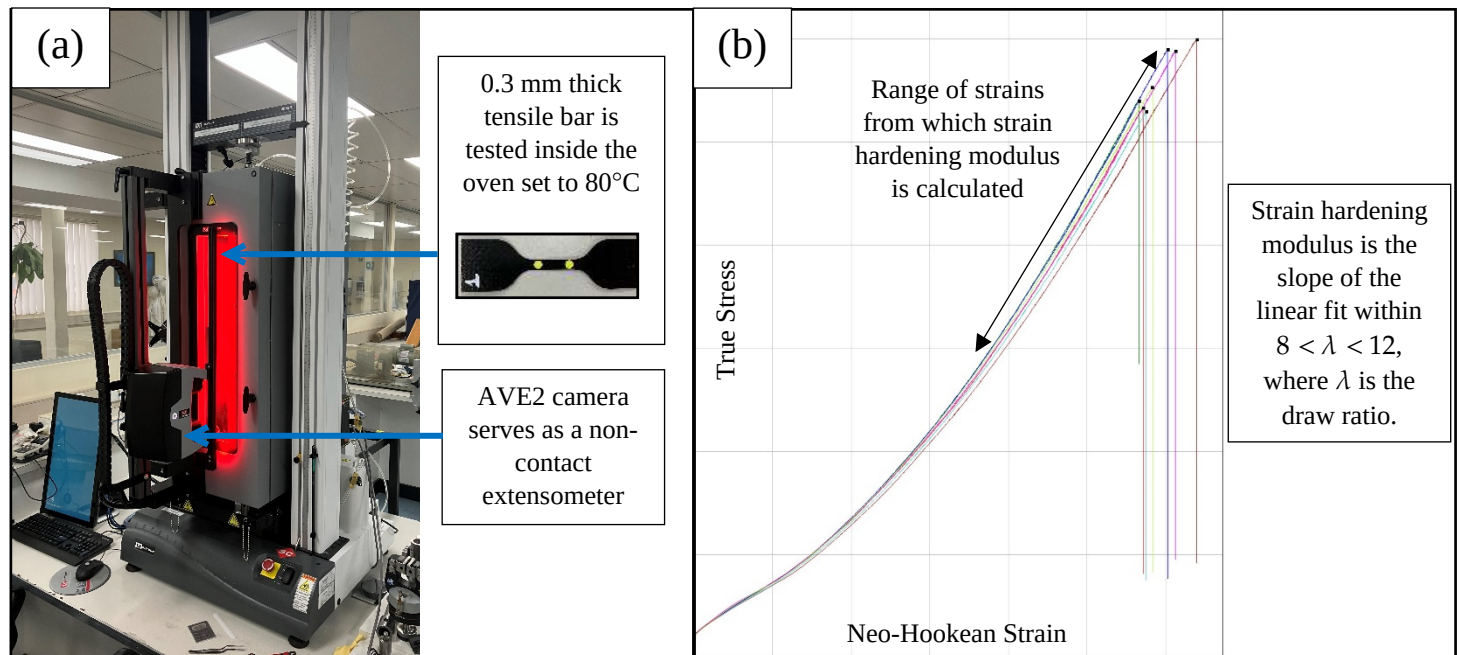


Figure 4: (a) photo of the strain hardening setup used to test pipe grade tensile samples. Due to the samples being 0.3 mm thick and exposed to high temperatures, a non-contact extensometer was more appropriate. Extension of the sample would be measured by tracking the distance between two yellow dots placed on the sample. (b) the gradient calculated from the true stress – Neo-Hookean strain for draw ratios within $8 < \lambda < 12$ gave the strain hardening modulus G_p .

Objectives and Motivation

Aim

- To establish a methodology for measuring the strain hardening (SH) modulus of pipe grade materials for research and development and quality control purposes.

Motivation

- An in-house methodology was required to assess the slow crack growth resistance within pipe grade materials, specifically PE100RC grades (RC = resistant to crack). Recently, SH tests have been identified as a standardized method to assess the crack resistance of PE100RC materials and has been introduced in EN/ISO pipe pressure standards, in addition to POP004 and POP016 standards.
- Advantages include being able to assess the performance of new crack resistant pipe grades (e.g. PE100 RCD), address any quality performance issues that may arise with customers, and reassure that the PE100RC grades sold to customers inherit the crack resistance properties as advertised. Significant costs are also saved due to no longer needing to rely on external laboratory testing for strain hardening data.

Results

Summary of methodology

- A tensile test is performed at elevated temperature (80 °C), from which a strain-hardening modulus (G_p) is derived from the stress-strain data for draw ratios (λ) within the range $8 < \lambda < 12$. G_p is proportional to the material's slow crack growth resistance.

Process to achieve strain hardening setup

- Calculations for the strain-hardening modulus based on the stress-strain curve were coded into the software used by the Instron, establishing a workflow that would be easy for the operator to follow when doing the test.
- A camera system was used as a non-contact extensometer and was configured to capture the full extension.
- Several iterations were required to obtain a precise and accurate sample preparation process, and multiple parameters (e.g. sample thickness, gauge length, camera distortions, and oven temperature) needed to be controlled to give results that were comparable to external laboratory testings.
- All testing and calibration procedures (e.g., camera and oven temperature calibrations) were appropriately documented within Qenos' laboratory database, with individuals in the laboratory appropriately trained.

Developing user interfaces to support QA testing and product releasing

Sept. 2023 – Jan. 2024 | QENOS

A particular batch is selected from the directory of results (e.g. #4024). After the first instance of getting data for a batch, the results and images are locally stored such that future retrievals of data is quicker.

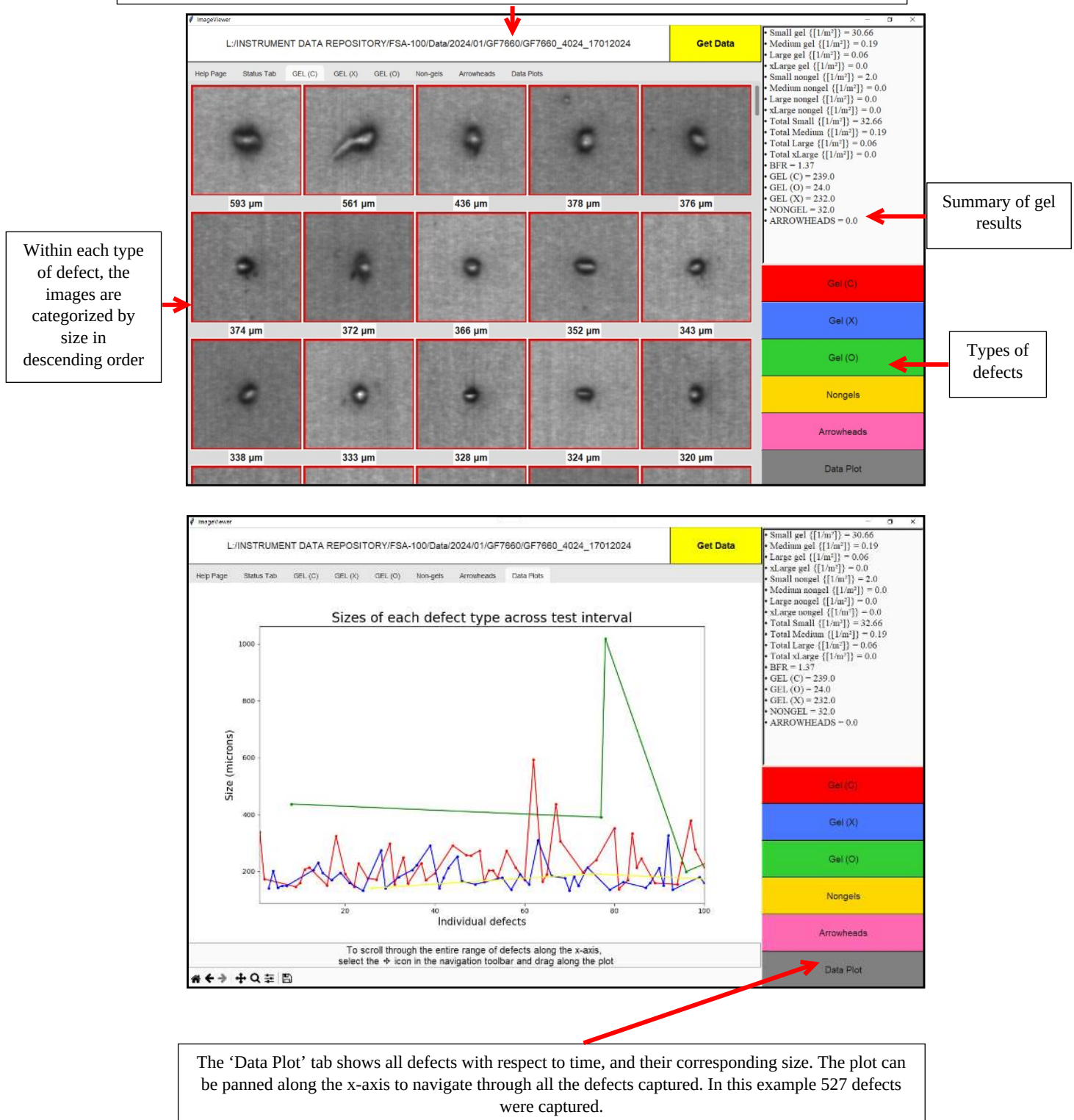


Figure 5: as part of regular QA testing for film products, each batch is run through a film extruder whilst a camera captures photographs of any defects that may appear. Each defect is characterised by type and size. The BFR metric is calculated based on the defect data, which is used by film testers to characterise film quality. A graphical user interface was made to remotely retrieve defect results (including their images), for which product integrity and operations personnel can use to assess the film.

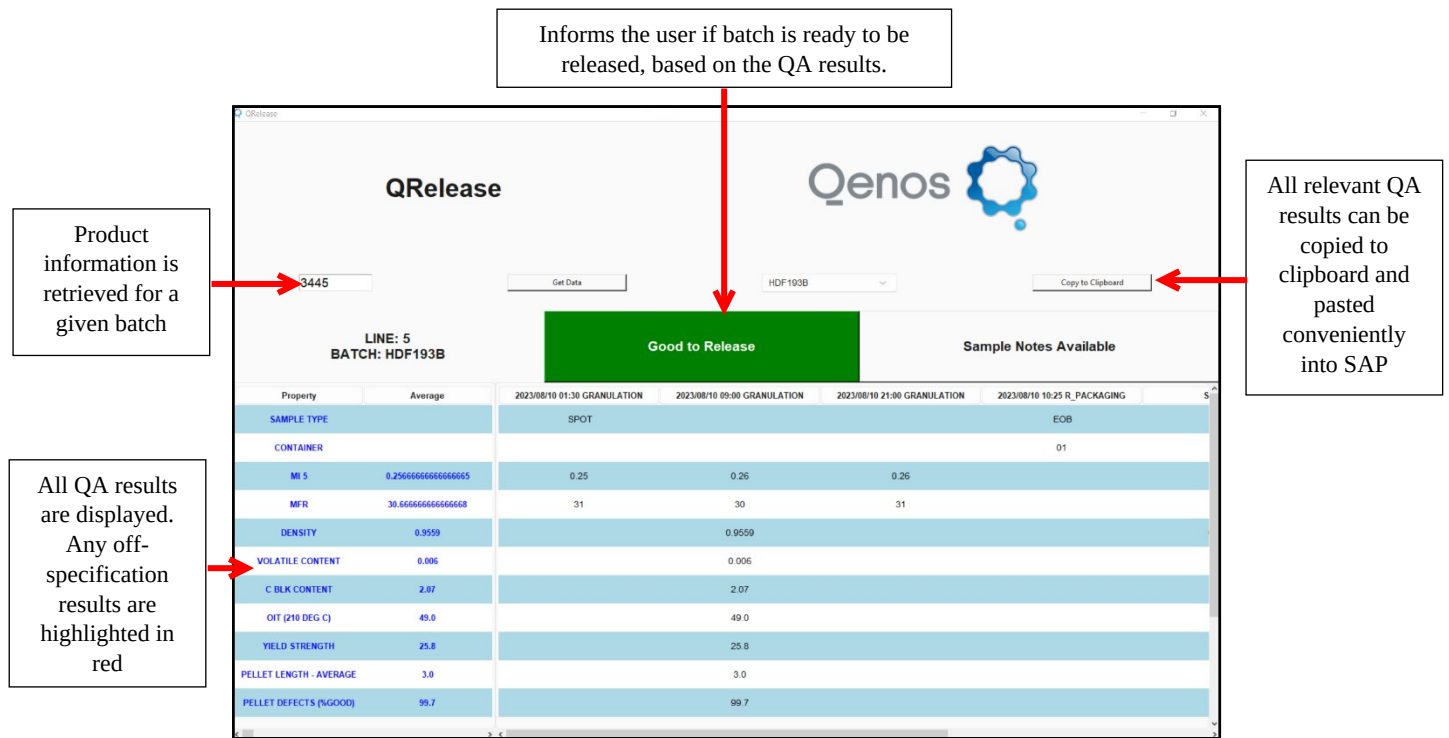


Figure 6: a graphical user interface was developed for product releasing, which summarises all the relevant QA results and product information for a given batch. Hence, this interface can conveniently provide all the information necessary to input into SAP for product release.

Objectives

Aims

- Interface for quality assurance:
 - Aim is to extract results from a camera system after QA gel testing and display the information in a way that is remotely accessible by technical service, product integrity, and operations.
- Interface for product releasers:
 - Aim is to extract product information and results from the plant's oracle SQL server and display the data for product releasers to have a summary of the relevant information for releasing.

Results

Interface for QA testing of extruded polyethylene film (Fig. 5):

- Results from quality assurance test of polyethylene film is exported as JPEG images, each containing a maximum of 32 defect photos. The images are ordered with respect to time. An excel file is also automatically generated detailing each defect's size and type.
- Computer vision is used to extract and sort the images into their respective defect type categories. Within each category, the images are sorted by size in descending order. Since the largest defects majorly affect the BFR (a metric used to quantify a film's quality), this makes it easier for QA testers and product integrity personnel to determine which defects are mostly impacting the film's quality.

Interface for product releasers (Fig. 6):

- Data from a SQL server is queried and tabulated in a way to summarise all the QA results and product information
- Results are colour coded to indicate to the product releaser whether the batch (product) is off-specification or requires further attention
- This provides releasers an efficient method to check whether all data results have been logged (and are in-specification) and ensure all product information is correct.

Final Year Engineering Project: Modelling Fe & Cr interactions in Fe-MIL-101 (Cr) derived catalysts

Mar. 2022 – Oct. 2022 | MONASH UNIVERSITY

Objectives and Motivations

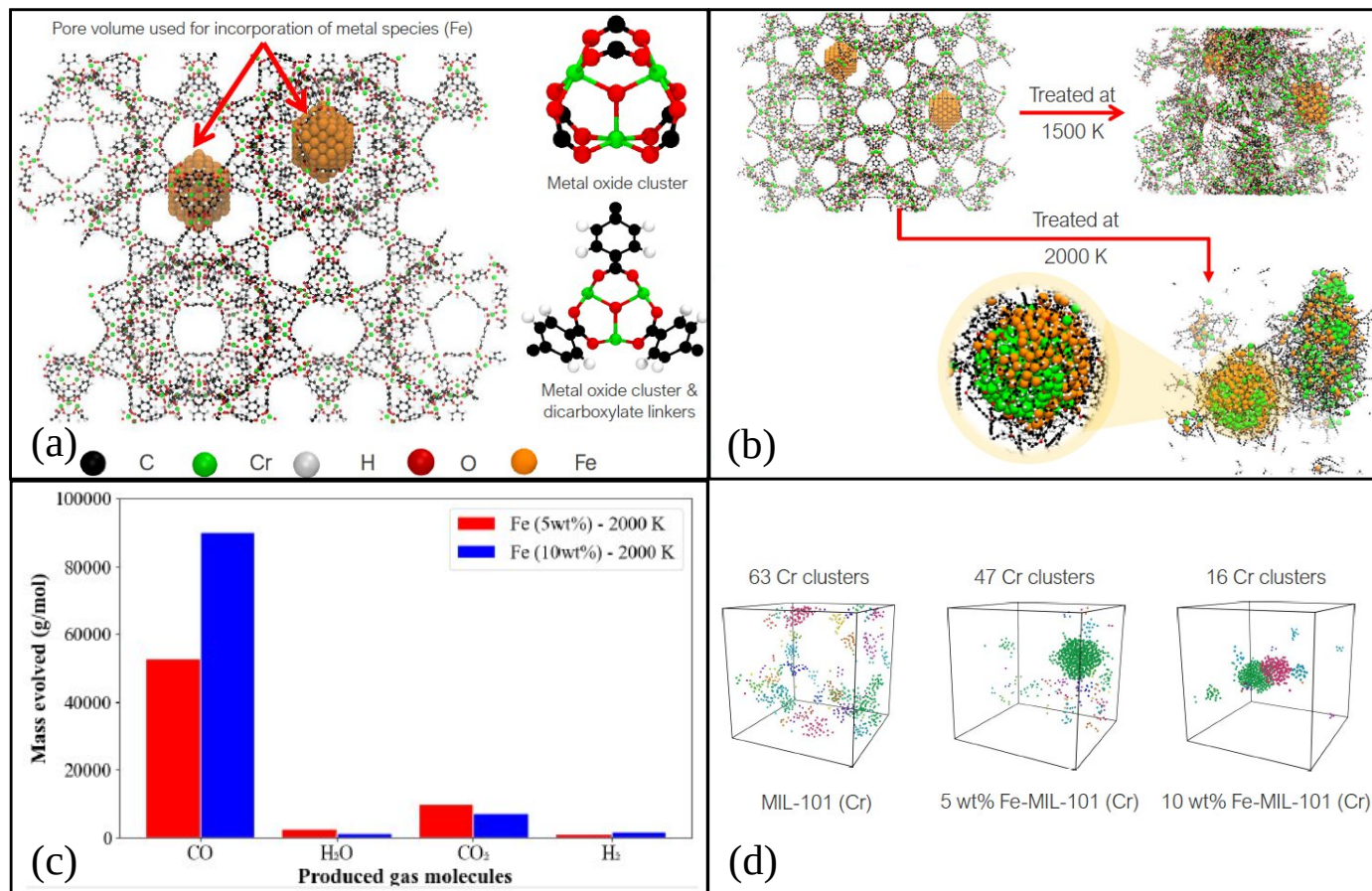


Figure 7: (a) structure of Fe-MIL-101 (Cr), where a series of Cr-O clusters are connected by 1,4-dicarboxylate linkers. In this example 10 wt% Fe is impregnated into the pores of MIL-101 (Cr). (b) thermal degradation of Fe-MIL-101 (Cr) is shown at two different temperatures. Fe atoms segregated to sintered Cr clusters at higher temperatures. (c) shows the gases evolved during thermal degradation. (d) the number of Cr clusters is shown for increasing Fe concentration.

Aims and motivations

- Metal organic frameworks (MOFs) are polymers with metal clusters connected via organic ligands
- Experiments show Cr and Fe mixed in MIL-101 (a type of MOF) contains high catalytic activity when thermally degraded:
 - Cr³⁺ enhances the catalyst's framework at high temperatures and Fe³⁺ enhances the MOF's catalytic activity (Fig. 7a)
- Thus, the aim was to study Fe-MIL-101 (Cr) derived materials at high temperatures using reactive molecular dynamics to understand both Fe and Cr chemistries and overall MOF morphology at the molecular level

Computational methodology

- MIL-101 (Cr) structures containing 5 and 10 wt% Fe were studied at high temperatures
- Due to the limited computational resources available, the MOFs were simulated at extreme temperatures but within feasible timescales (nanoseconds) to accelerate the thermal transformations.
- MOFs were simulated under two different temperature regimes: 1500 and 2000 K.

Results

- At high temperatures Fe atoms segregated to sintered Cr clusters, establishing long-range order of Cr-Fe interactions (Fig. 7b)
- Higher Fe concentrations drive catalytic reaction of CO (Fig. 7c)
- Higher Fe concentrations promote Cr core shell formation (Fig. 7d). Fe₂O₃ and Fe₃O₄ molecules formed on Cr surfaces (which was confirmed from radial distribution analysis). These oxides act as the main active sites for catalytic reactions.
- Thus, by studying the impact of Fe concentrations and temperature on Fe-Cr interactions in Fe-MIL-101 (Cr), we developed an understanding of how we can tune the properties of these thermally degraded structures for various catalytic applications.

Composite manufacturing and Finite Element Analysis of a racing fairing

June 2021 – Nov. 2022 | MONASH UNIVERSITY

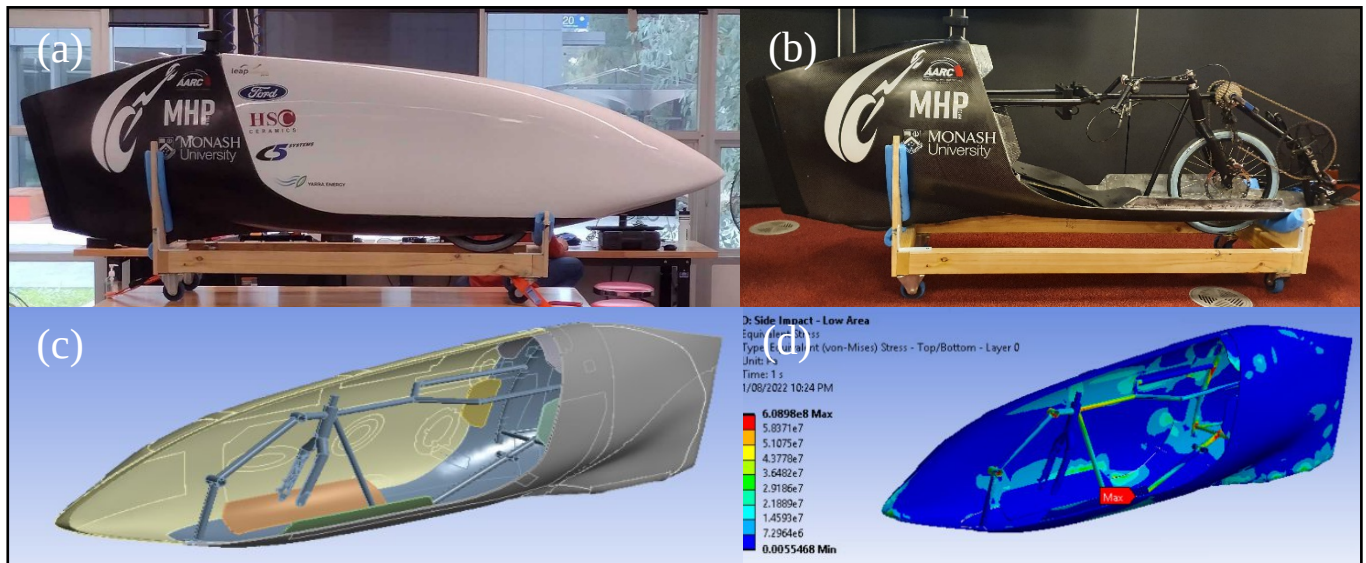


Figure 8: (a) vinyl-wrapped carbon-fibre fairing integrated with chassis. (b) front fairing removed to reveal chassis and drivetrain. (c) computer aided design of the fairing and chassis. (d) deformations and equivalent stresses are shown under a given static load.

Objectives and Motivation

Aims

- The primary objective was to manufacture a carbon-fibre fairing for a competitive racing cycle (Figs. 8a and 8b) using a combination of various composite manufacturing techniques: wet-layups, vacuum bagging, and resin-infusion.
- Finite element analysis software (ANSYS-mechanical) was also used to perform static simulations of the carbon-fibre fairing under relevant loading scenarios for safety-critical analysis.

Motivation

- The team is actively involved in human powered vehicle (HPV) competitions, and thus a weight saving, fatigue resistant, mechanically strong fairing was desired, leading to the need for a composite derived racing fairing.

Manufacturing Process

- A gel-coated CNC machined male mould was laid with fibreglass reinforcement to produce a female mould forming the basis for the fairing's manufacturing.
- Both fairing halves were laid with carbon fibre, Kevlar, and PVC core submerged in epoxy resin.
- The resulting halves from resin infusion were bonded and then carved to form the front and rear sections.

Results

Finite Element Analysis (FEA) results

- A side-crash scenario was statically simulated using a point load situated at the side of the fairing, from which the corresponding deformation and equivalent stresses (Fig. 8d) were measured (maximum deformation and stresses experienced were 8.2 mm and 609 MPa respectively)

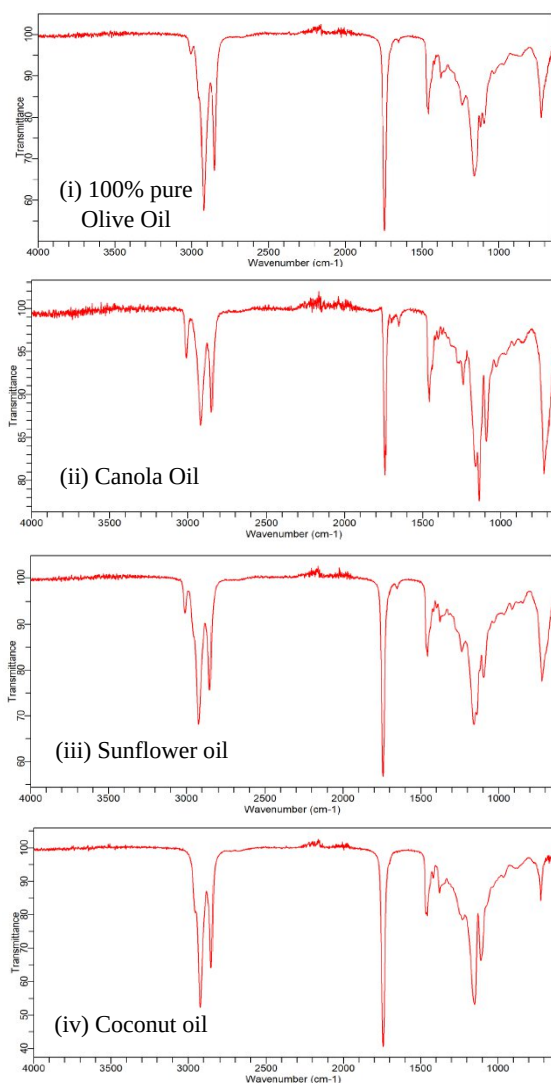
Competition results

- Upon testing at Human Powered Vehicle (HPV) competition in Nevada, the racing cycle achieved a peak speed of 116 km/h, setting the Australian record for the fastest two-wheeled HPV.

Chemistry project: detection of adulteration practices in commercial oils

Sep. 2019 – Nov. 2019 | MONASH UNIVERSITY

(a) FTIR Analysis



(b) Raman Analysis

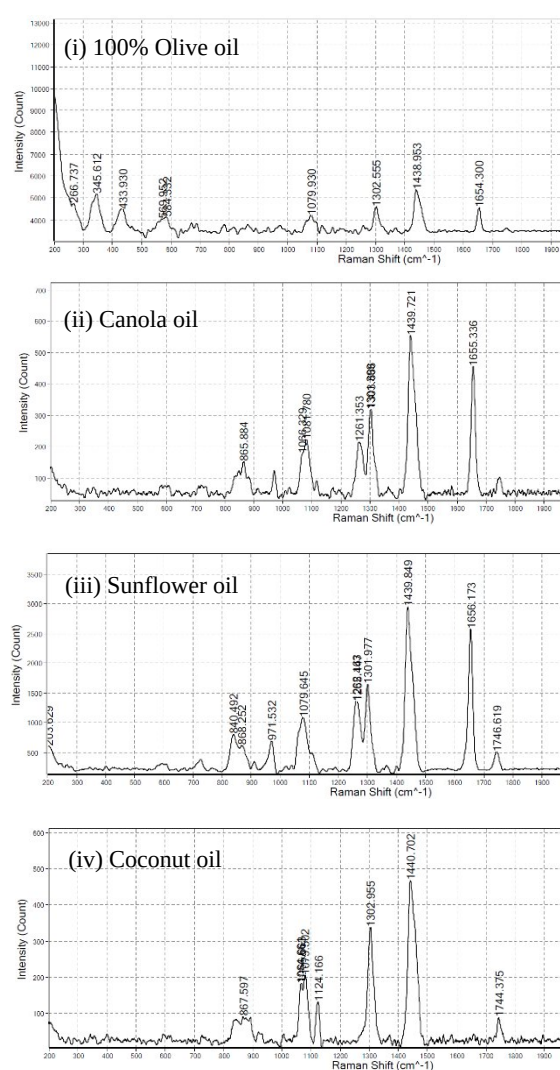


Figure 9: summary of (a) FTIR and (b) Raman spectra of olive oil and common adulterants: canola, sunflower, and coconut oils

Objectives and Processes

Aim

- Aim was to determine whether the detection of oil adulteration practices in commercial olive oil brands can be accomplished using Fourier-transformed infrared (FTIR) and Raman spectroscopy methods

Motivation and Method

- Commercial olive oils adulterated with coconut, sunflower, or canola oils have been previously publicly reported and thus both FTIR and Raman spectroscopy methods were selected in an attempt to distinguish between the oils by their level of unsaturation and fatty acid composition
- FTIR and Raman spectra were obtained of all adulterants, in addition to three unknown adulterated olive oils samples supplied by Monash's School of Chemistry

Results

- The degree of unsaturation was used to distinguish the similar FTIR and Raman spectra of adulterants and 100% pure olive oil (shown in Figs. 9a and 9b)
- 100% olive oil was highly monosaturated (mainly comprised of oleic acid), with sunflower and canola oils being highly polyunsaturated (linoleic acid) and monounsaturated (oleic acid) respectively, and coconut oil being highly saturated
- Based on this information, qualitative and quantitative analysis of adulterated olive oil samples was performed and successfully concluded the adulterants evident in each of the three unknown olive oil samples, demonstrating the capabilities of using FTIR and Raman spectroscopy to detect the purity of commercial olive oil samples